

# Accurate continuous action and gesture recognition method based on skeleton and sliding windows techniques

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**Abstract**—Vision-based action recognition with the main aim to determine the action class from image sequence has emerged recently thanks to its potential applications. Numerous methods have been proposed for action recognition. However, majority of them mainly focus on isolated action recognition with the assumption that videos are segmented in different clips or shots, each of them contains an image sequence of one sole action. Different from isolated action recognition, continuous action recognition involves recognizing human actions from a continuous stream of input data without requiring pre-segmentation into discrete action instances. In this paper, we propose a method for continuous action recognition. The proposed method incorporates sliding window technique and a light weight action classification model named DDNet. We evaluate the proposed approach on several benchmark datasets. Experimental results show that the proposed method the accuracy of 0.95, 0.97 and 0.71 for IPN, UOW and InHARD datasets respectively on isolated action recognition task. On continuous recognition task, it has achieved a performance of 0.57, 0.76 and 0.33 on IPN, UOW and InHARD datasets.

**Index Terms**—skeleton-based action recognition, continuous action recognition

## I. INTRODUCTION

The ability to recognize and understand human actions in real-time has important applications in various domains, including surveillance, human-robot interaction, and healthcare. In surveillance, distracting moments of drivers, and illegal house intrusion can be detected [1], [2]. In human-robot interaction, gesture and action recognition can enable robots to respond to human commands and intentions, facilitating natural and intuitive communication [3] while in healthcare domain, action recognition can be used to monitor and assist patients through physical rehabilitation [4].

Recent advances in deep learning have enabled significant progress in action recognition, leveraging approaches, including Convolutional Neural Networks

(CNNs), Recurrent Neural Networks (RNNs), and graph-based models. These methods have demonstrated encouraging outcomes in the identification of diverse human actions within video sequences.

However, majority of existing work focus on isolated action recognition with the assumption that videos are segmented in different clips or shots, each of them contains an image sequence of one sole action. Different from isolated action recognition, continuous action recognition involves recognizing human actions from a continuous stream of input data without requiring pre-segmentation into discrete action instances.

Compared with isolated recognition, recognizing continuous actions poses a notably greater challenge as it has to tackle to variation of actions' length, unclear boundary between actions.

In this paper, we introduce a novel approach designed for continuous action recognition. The proposed method incorporates sliding window technique and a light weight action classification model named DDNet [5]. We assess the performance of our approach on various benchmark datasets and conduct a comparative analysis with state-of-the-art approaches, demonstrating its effectiveness and robustness in continuous action recognition tasks.

The rest of this paper is organized as follows: Section 2 provides a review of the related work in human action recognition. Section 3 describes the detailed information of the proposed approach. Section 4 shows our experiments on three different datasets of continuous action and gesture recognition and analyze in detail the experimental results. Conclusion and future works are illustrated in Section 5.

## II. RELATED WORKS

In this section, we provide an overview of the related work on action recognition, highlighting the

key contributions and limitations of existing methods. The study on video-based action recognition can be categorized into two fundamental groups: isolated and continuous action recognition.

Isolated action recognition takes split videos that contain only a single instance of action as inputs and predicts the action label. Many approaches have been proposed for isolated action recognition. Some authors used CNNs to capture local spatial patterns and temporal dynamics, thereby effectively modeling the complex motion characteristics of actions such as [6]. Due to the limitation of the single-frame image data input, some authors proposed to use 3D CNN to learn the temporal information for action recognition including different fusion approaches, spatio-temporal attention mechanisms, and dual-stream 3D-CNN [7], [8]. In addition, the method based on Long Short-Term Memory (LSTM) and CNN that have been proposed to recognize actions quickly and accurately [9]. Nayan *et al.* [10] proposed a combined RNN that uses CNN to extract features for a bidirectional LSTM (BD-LSTM). While CNN, 3D CNN and LSTM based methods mainly work on RGB sequence, different GCN models have been proved to be robust for skeleton-based action recognition.

Recently, some methods have been proposed to address continuous action recognition from RGB [11] or skeleton sequences [12]. These methods built a detector to determine whether an images sequence contains an action or not and a classifier that aims to identify the action class. Building a detector is tedious task as it requires to prepare a training set that consists of image sequences for non action/non gesture. Moreover, some methods could not guarantee real time processing. In [13], the authors propose a model consisting of a Spatio-Temporal GCN and the sliding window method for continuous action recognition from skeleton sequence. A sliding window of a predetermined size was applied to a continuous stream of skeleton data, and it was shifted as time advanced. This method allows to accelerate the recognition speed, however, GCN is still complex model that requires computational resources. In [12], a light model has been introduced for continuous hand gesture recognition. However, the model is designed for IPN dataset and could not work on other datasets.

### III. PROPOSED METHOD

#### A. Sliding Window

The proposed strategy, which is shown in Fig. 1, is divided into two stages: training and inference. In the training step, a light weight model based on DDNet is trained on isolated sequences of different actions. Then, in the inference step, sliding window technique

is incorporated to recognize the dynamic gestures from continuous stream of video. The window size denoted  $ws$  needs to be set to a fixed number same as the input of DDNet. The number of frames in the sliding window is will be chosen for each dataset depending on the frame rate and the length of gestures. This window strides every 1 frame and returns the predicted label for the middle frame of the window.

#### B. DDNet

DDNet (a Double-Feature Double-Motion network) is a light weight model proposed for skeleton-based human action recognition [5]. Although being a light weight model, DDNet and its variants provides state-of-the-art results on different datasets [2].

In Fig. 2, we conducted a performance comparison of recognition models using the MSRAAction dataset [14] in term of both FLOPs and accuracy. We can observe that DD-Net results a good recognition rate while requires less computational time compared to GCN model.

DDNet consists of three branches that aims at extracting both spatial and temporal features from skeleton sequence. There branches are Slow motion, Fast motion and Joint Collection Distances (JCD). Each branch contains three CNN layers. Slow and Fast motions are designed to capture the movement of the actions while JCD is dedicated to spatial information.

Denote  $S^{t+1}$  and  $S^{t+2}$  are the skeleton of one frame and two frames behind the skeleton at frame  $t$   $S^t$ ,  $F_{slow}^t$  and  $F_{fast}^t$  that are denoted as the Slow Motion the Fast Motion and are computed as follows:

$$F_{slow}^t = S^{t+1} - S^t, t \in 1, 2, 3, \dots, T-1; \quad (1)$$

$$F_{fast}^t = S^{t+2} - S^t, t \in 1, 3, \dots, T-2; \quad (2)$$

Concerning JCD, it is extracted from a Euclidean distance matrix. Each element of the matrix represents the Euclidean distance between a pair of joints in the same frame. At frame  $t$ , JCD feature can be computed by using the following equation.

$$JCD^t = \begin{bmatrix} \|\vec{j_2^t j_1^t}\| & & \\ \vdots & \ddots & \\ \|\vec{j_N^t j_1^t}\| & \dots & \dots & \|\vec{j_N^t j_{N-1}^t}\| \end{bmatrix} \quad (3)$$

where  $\|\vec{j_k^t j_l^t}\|$  ( $k \neq l$ ) is the Euclidean distance between two joints  $j_k^t$  and  $j_l^t$  of the skeleton  $S^t$ .

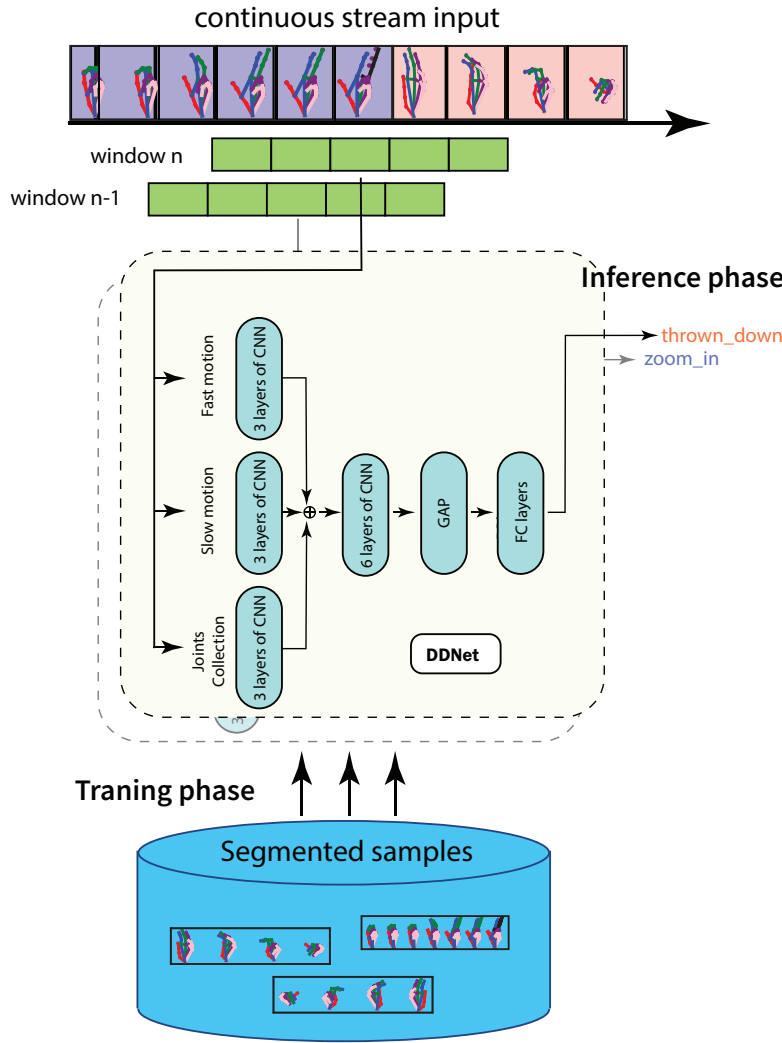


Fig. 1: Proposed method for skeleton-based continuous human action recognition

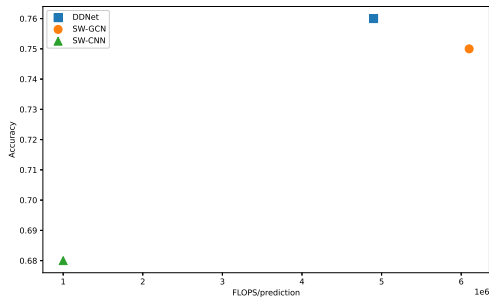


Fig. 2: Comparison of recognition models on MSRAAction dataset [14]

TABLE I: Accuracy obtained by the proposed method on 3 datasets for isolated classification task.

|     | IPN  | UOW  | InHARD |
|-----|------|------|--------|
| Acc | 0.95 | 0.97 | 0.71   |

TABLE II: Comparison of the proposed method with state-of-the-art methods on three benchmark datasets

|                 | IPN         | UOW         | InHARD      |
|-----------------|-------------|-------------|-------------|
| SW-CNN [13]     | -           | 0.68        | -           |
| SW-GCN [13]     | -           | 0.75        | -           |
| Proposed method | <b>0.57</b> | <b>0.76</b> | <b>0.33</b> |

## IV. EXPERIMENTAL RESULTS

### A. Datasets

We conducted our experiments on the following continuous gesture datasets, which is widely used for evaluating continuous action recognition algorithms. **IPN** [11]: This dataset includes over 4,000 gesture samples and 800,000 RGB frames from 50 different individuals. 13 different static and dynamic gestures were created to focus on interactions with touchless screens. The main characteristic of the dataset is that the continuous gestures are performed without tran-

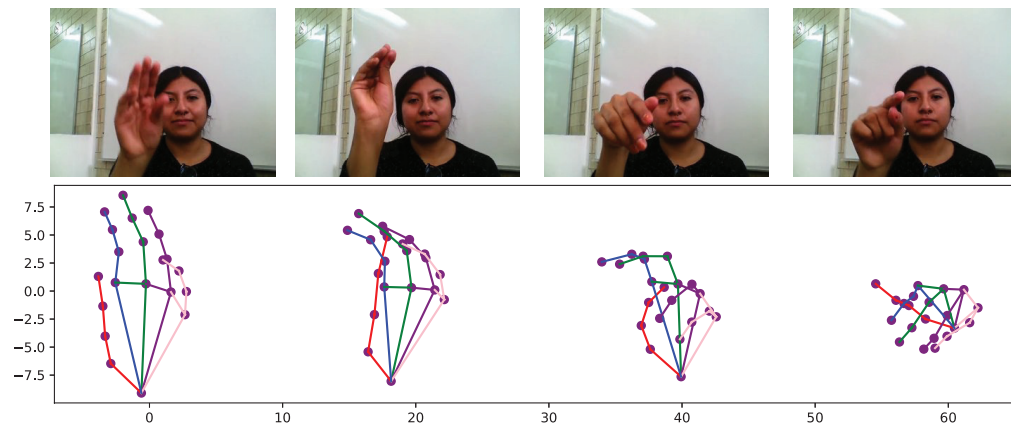


Fig. 3: RGB images sequence and its corresponding skeleton sequence of "Throw down" action in IPN dataset

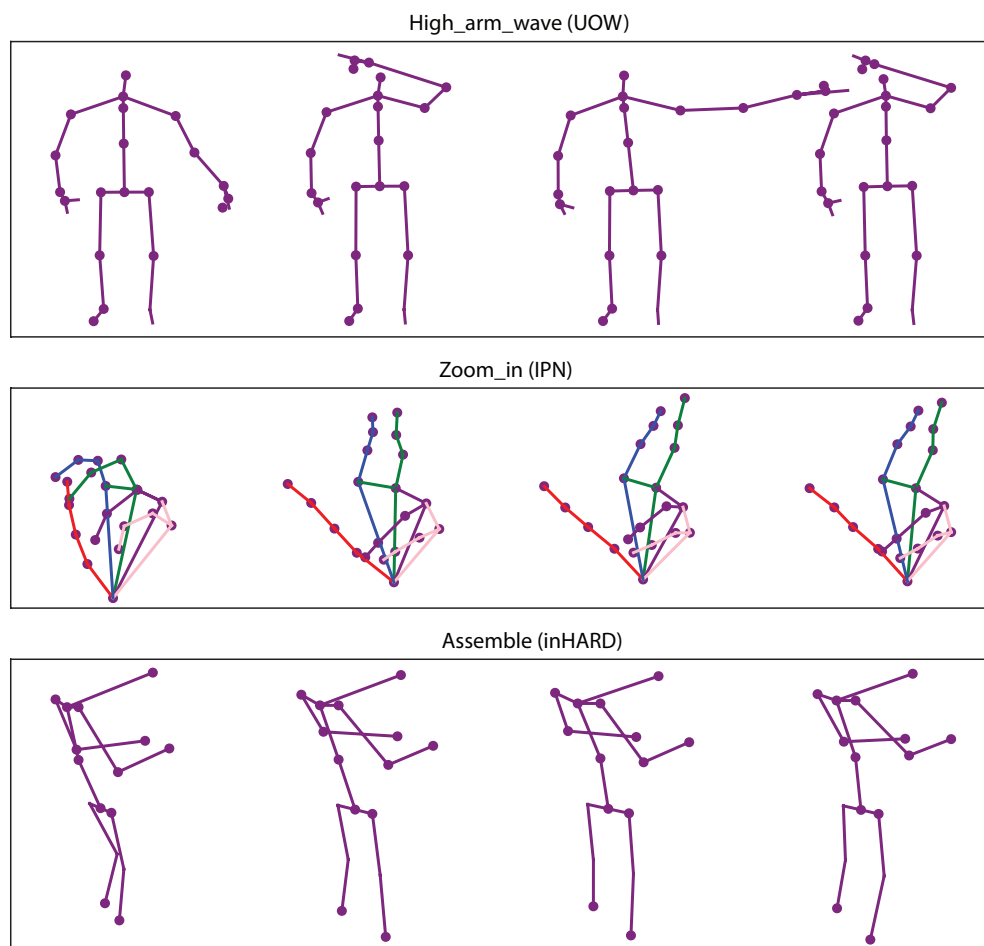


Fig. 4: Example of skeleton sequences of UOW, IPN and InHARD datasets

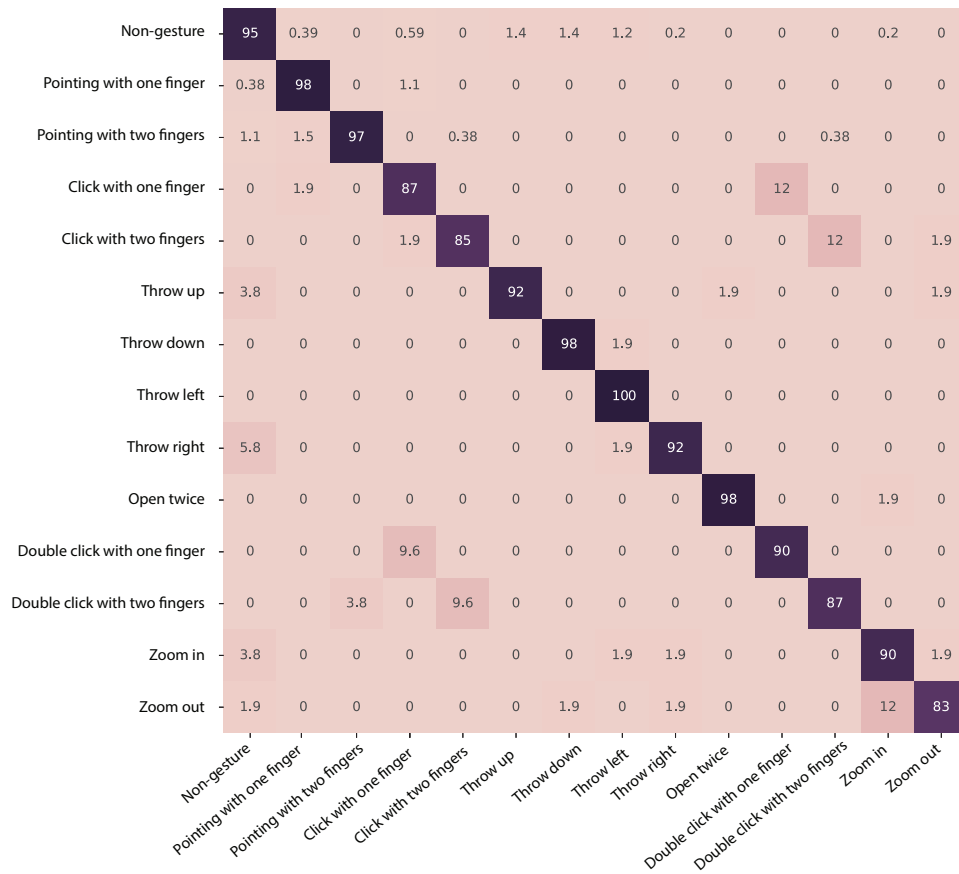


Fig. 5: Isolated action classification on IPN dataset

sition states and non-gesture actions are performed by subjects using natural hand movements. Gestures were captured in about 30 different settings with various backgrounds and lighting conditions to introduce real-world variation. Since the skeleton information is not available on this dataset, we estimate hand joint landmarks from provided video by using Mediapipe [15], an open-source cross-platform framework developed by Google that provides a pipeline for building real-time applications.

**UOW:** [16] The UOW Online Action3D Dataset contains action sequences of 20 actions. These actions are defined in MSR Action3D Dataset. The dataset were captured using Microsoft Kinect V.2 and comprises videos of 20 subjects. Each subject performed one action from 3 to 5 times before performing 20 actions continuously in a random order. The dataset has two subsets: a subset of continuous action sequences can be utilized for evaluation of continuous action recognition methods while the subset of repeated action sequences will be used for isolated action recognition or training purposes.

**InHARD (Industrial Human Action Recogni-**

**tion Dataset):** [17] has recently released for human-COBOT interaction context. This dataset contains over 4800 action samples that perform 13 different industrial action classes named meta-action. The authors use a motion sensor system to capture the human skeleton at 120 Hz. Figure 4 shows example of skeleton sequence of these datasets. It is worth to note that the number of joints for IPN, UOW and InHARD datasets are 21, 25 and 17 respectively.

#### B. Implementation details and evaluation metrics

The size of the window depends on the frame rate and gesture length of each dataset. We set the window size of UOW and IPN is 60 and of inHARD is 240. All experiments were conducted on a PC with an Intel Core i7 8750H processor.

To assess the performance action recognition system, we employ the video-wise accuracy for isolated recognition and frame-wise accuracy for continuous recognition. Frame-wise accuracy compares the predicted label vs the ground-truth label at frame level.

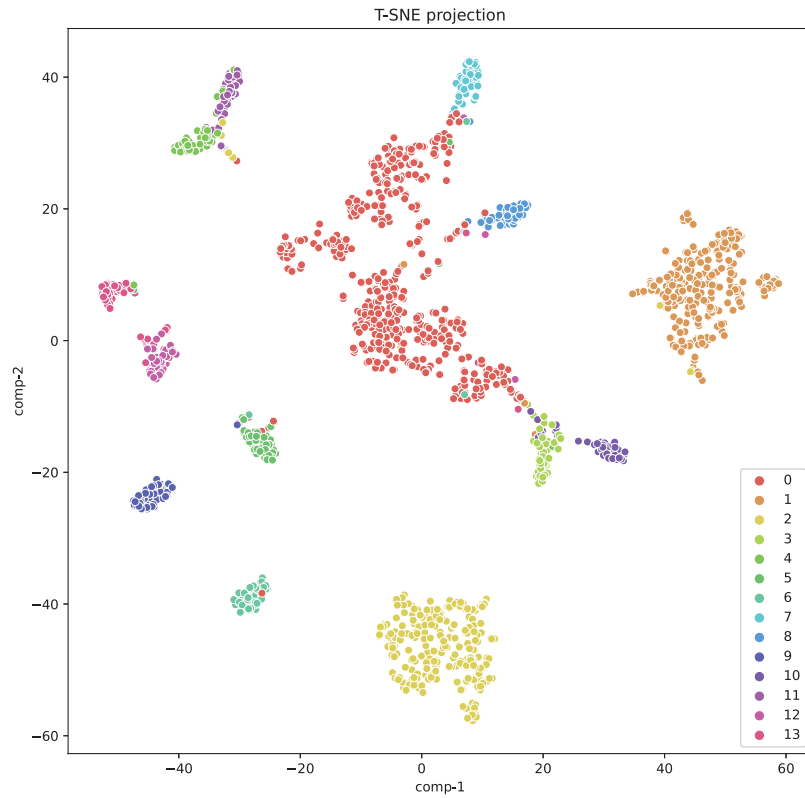


Fig. 6: Feature space learned from DDNet model for IPN dataset

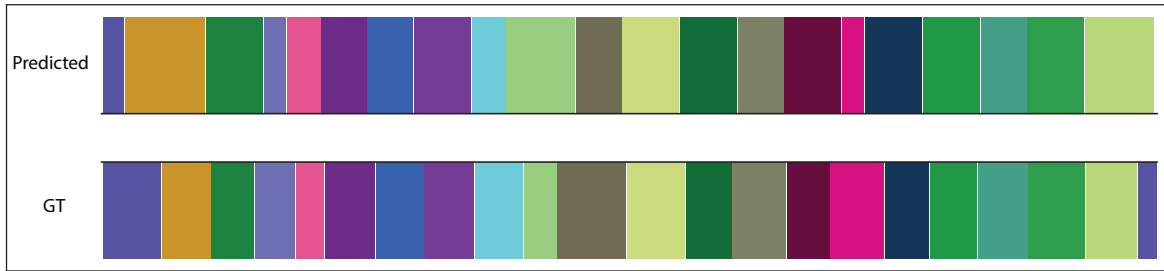


Fig. 7: Illustration of continuous action recognition on a sequence of UOW dataset. The first line represents the results of the proposed method while the second line shows the corresponding ground-truth. Each action label is represented by an identical color.

### C. Results

As the proposed method consists of two main components: the sliding window technique and an action classifier based on DDNet. We evaluate first the performance of action classifier. Table I shows the accuracies obtained for three benchmark datasets. The accuracies obtained for IPN, UOW and InHARD are 0.95, 0.97 and 0.71 respectively. We can observe that the model based on DDNet could distinguish different actions with high accuracy. Among three datasets, the performance obtained for InHARD is lowest. The main reason is that in InHARD contains 74 actions belonging to 13 meta-actions, each type of

meta-action has many variations of actions, therefore, it is hard to achieve the high accuracy in classification task. For IPN dataset, even with skeleton extracted by off the shelf method [15] and the high inter-similarity between action classes, the method still could recognize very well the actions. Figure 3 illustrates the RGB sequence and its corresponding skeleton sequence for "Throw down" action of IPN dataset. Figure 5 shows the confusion matrix obtained for IPN while Fig. 6 shows the feature distribution of 13 actions in IPN dataset using t-SNE method. We can observe that the "Click" and "Double click" actions get a lot of confusions between 1 and 2

fingers. This can be explained by the similarity in the movement. In some cases, 1 of 2 fingers can be obscured leading Mediapipe only estimating 1 finger. Table II compares the performance of the proposed method with CNN and GCN based methods in [13] on three benchmark datasets: IPN, UOW, and InHARD. The two first row represents results of the SW-CNN, the SW-GCN method [13], which has achieved a performance of 0.68, 0.75 in term of frame-wise accuracy on the UOW dataset. The last row represents the proposed method, which has achieved a performance of 0.57 on the IPN dataset, 0.76 on the UOW dataset, and 0.33 on the InHARD dataset.

Figure 7 illustrates recognition for a continuous sequence in UOW dataset. The first line represents the proposal's results whereas the second line shows the corresponding ground-truth. Each action label is represented by an identical color. From this figure, we can observe that the capability of the proposed method to accurately localize and recognize different action instances from input sequence. However, for some actions such as the two first segments present the transition from non-gesture to "pick up & throw" gesture. Because the sliding window returns the label of the middle frame, in this case, skeletons in the window contains both non-gesture frame and the starting frame of "pickup" gesture. Therefore, it is recognize as pick up gesture.

## V. CONCLUSION AND FUTURE WORKS

In this paper, a method for continuous action recognition from a skeleton sequence has been proposed. The proposed method incorporates sliding window technique and a light weight action classification model. Based on our experiments conducted on three benchmark datasets, the proposed method achieved remarkable accuracy scores of 0.95, 0.97, and 0.71 for the IPN, UOW, and InHARD datasets, respectively. These results pertain specifically to the isolated action recognition task. On continuous recognition task, it has achieved a performance of 0.57, 0.76 and 0.33 on IPN, UOW and InHARD datasets. Although the proposed method obtains promising results for isolated recognition task, there is still room for improvement for continuous recognition especially when working with challenging dataset such as InHARD.

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